Chemistry Topic 3 — Metals, Non-Metals and Chemical Reactions

Sources baked in: NCERT Class 8, 9 & 10 (Metals and Non-metals; Chemical Reactions and Equations), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Highly recurring questions on German Silver composition (0% Ag), Galvanization (Zinc coating), Amalgam exceptions (Iron does not form amalgams), Graphite stability vs Diamond hardness, Aqua Regia ratio (3:1 HCl:HNO3), and Ore identification (Cinnabar, Bauxite, Pitchblende, Galena). **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Critical Minerals Mission & Lithium Reserves**	Geological Survey of India (GSI) exploration of **Lithium** reserves in Reasi (Jammu & Kashmir) and Degana (Rajasthan). Lithium (Li, $Z = 3$) is the lightest metal and lowest-density solid element ($0.534\ \text{g/cm}^3$), with the highest negative standard reduction potential ($-3.04\ \text{V}$), making it the bedrock of EV lithium-ion batteries.
2024	**UPPCS Prelims 2024**	Question on **German Silver composition**: Highlighted the recurring trap that German Silver contains 0% **Silver** ($Cu \approx 50\%, Zn \approx 30\%, Ni \approx 20\%$). Named solely for its silvery-white metallic luster.
2023	**Noble Prize in Chemistry & Carbon Materials**	Advancements in **Graphene** (single atomic layer of sp^2 carbon) and Carbon Nanotubes (CNTs) for supercapacitors and aerospace composites: Graphene exhibits ballistic electrical conductivity and tensile strength > 100 times higher than structural steel.
2022	**UPPCS Prelims 2022**	Core thermodynamic trap: **Graphite** is the thermodynamically most stable allotropic form of carbon at standard temperature and pressure ($298.15\ \text{K}, 1\ \text{bar}$), with standard enthalpy of formation $\Delta_f H^\circ = 0\ \text{kJ/mol}$ (Diamond has $\Delta_f H^\circ = +1.9\ \text{kJ/mol}$).
2020	**UPPCS Prelims 2020**	Characteristic matching of metals: Sodium (soft, cut with a knife), Mercury (liquid metal), Silver (highest electrical conductivity), and Lead (poorest thermal conductor among common commercial metals).

Detailed Theory Notes

3.1 Physical Properties & Unique Tags of Metals and Non-Metals

1. Mechanical Properties

- **Malleability**: The property that allows a material to be hammered, rolled, or beaten into ultra-thin sheets without fracturing. Gold (Au) is the most malleable element known, followed by Silver (Ag) and Aluminium (Al).
- **Ductility**: The mechanical property of being drawn out into fine longitudinal wires under tensile stress. Gold is the most ductile metal: $1\ \text{g}$ of gold can be drawn into a wire $2\ \text{km}$ long.
- **Hardness vs. Softness**: Most transition metals (Tungsten, Chromium, Iron) exhibit extreme hardness due to overlapping d-orbitals forming strong metallic bonds. In stark contrast, alkali metals (**Sodium, Potassium, Lithium**) have only one valence s-electron and exceptionally large atomic radii, resulting in weak metallic bonding; they can be effortlessly cut with a butter knife.
- **Sonority**: The ability of a metal to produce a deep, ringing acoustic sound when struck, caused by rapid delocalized electron wave dissipation. Used in bells and musical wires (Bronze, Bell Metal).

2. Thermal and Electrical Conductivity

- **Conduction Mechanism**: Metals contain a "sea" of free, delocalized conduction electrons. When an electric potential or temperature gradient is applied, these electrons undergo directional drift, carrying thermal and electrical energy at high velocities.
- **Conductivity Sequence**:

$$\text{Electrical Conductivity: } Ag > Cu > Au > Al > W > Fe$$

  Although Silver is the superior conductor, **Copper and Aluminium** are preferred for domestic and industrial wiring due to cost efficiency and lower density.
- **Thermal Anomalies**: Lead (Pb) and Mercury (Hg) have the lowest thermal conductivity among common metals, while Diamond (a non-metal) has a thermal conductivity 4–5 times higher than Copper due to phonon lattice vibrations.

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Densest Element:   Osmium (Os, 22.59 g/cm³)
Lightest Metal:    Lithium (Li, 0.534 g/cm³)
Liquid Metal:      Mercury (Hg, m.p. -38.8°C)
Liquid Non-Metal:  Bromine (Br₂, reddish-brown)
Highest m.p. Metal: Tungsten (W, 3422°C)
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3.2 Carbon Allotropes & Thermodynamic Stability

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                                ┌── Diamond: sp³, tetrahedral 3D network; hardest; insulator;
                                ├── Graphite: sp², planar hexagonal layers; electrical conduc
Carbon Allotropes (Pure Carbon) ─┤
                                ├── Fullerenes: C₆₀ buckyball; 20 hexagons + 12 pentagons; ca
                                └── Graphene: Single 2D atomic sheet; ballistic conductivity;
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1. Diamond

- **Structure**: Each carbon atom is sp^3 hybridized and covalently bonded to 4 neighboring carbon atoms in a rigid, three-dimensional tetrahedral lattice with a uniform $C-C$ bond length of $1.54\ \text{\AA}$.
- **Properties**: Hardest known natural substance (10 on Mohs hardness scale); density $3.51\ \text{g/cm}^3$; high refractive index (2.42) and small critical angle (24.4°), causing brilliant total internal reflection.
- **Electrical & Thermal Behavior**: Because all 4 valence electrons are tightly locked in localized covalent σ-bonds, diamond has no free electrons and is a **complete electrical insulator**. However, its rigid lattice allows high-frequency phonon vibrations, making it an **exceptional conductor of heat**.

2. Graphite

- **Structure**: Each carbon atom is sp^2 hybridized and covalently bonded to 3 neighboring carbon atoms in a flat hexagonal planar ring ($C-C$ bond length $1.42\ \text{\AA}$). These planar layers are stacked parallel to one another at a distance of $3.35\ \text{\AA}$ and held together only by weak van der Waals dispersion forces.
- **Properties**: Soft, greasy, and slippery; layers slide over one another under shear stress, making graphite an ideal **solid dry lubricant** for high-temperature machinery where liquid oil would decompose.
- **Electrical Conduction**: The fourth valence electron of each carbon atom occupies an unhybridized p-orbital that overlaps sideways to form delocalized π-bonds across the hexagonal sheets. These free electrons move along the sheets, making graphite a **good conductor of electricity** (used as electrodes in dry cells and electric arc furnaces).
- **Thermodynamic Ground State**: In chemical thermodynamics, **Graphite is the most stable allotrope of carbon** under standard ambient conditions ($298.15\ \text{K}, 1\ \text{bar}$). By international IUPAC convention, the standard molar enthalpy of formation of graphite is defined as zero ($\Delta_f H^\circ\ [\text{graphite}] = 0\ \text{kJ/mol}$), whereas for diamond it is $+1.9\ \text{kJ/mol}$. Diamond is technically thermodynamically metastable and converts spontaneously—though infinitely slowly—into graphite.

3. Fullerenes & Graphene

- **Buckminsterfullerene (C_{60})**: Discovered in 1985 by Harold Kroto, Robert Curl, and Richard Smalley (Nobel Prize in Chemistry, 1996). Contains 60 carbon atoms organized in a spherical geodesic cage consisting of **20 hexagonal rings and 12 pentagonal rings**. No two pentagons share an edge (Isolated Pentagon Rule).
- **Graphene**: Isolated in 2004 by Andre Geim and Konstantin Novoselov at the University of Manchester using mechanical exfoliation ("Scotch tape method", Nobel Prize in Physics, 2010). Graphene is a single atom-thick two-dimensional sheet of sp^2 carbon atoms arranged in a honeycomb lattice with ballistic electronic mobility, extreme thermal conductivity, and tensile strength over $130\ \text{GPa}$.

3.3 The Reactivity Series & Chemical Behavior of Metals

The **Electrochemical / Activity Series** ranks metals in descending order of their standard oxidation potentials (tendency to lose electrons in aqueous solution):

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K  >  Na  >  Ca  >  Mg  >  Al  >  Zn  >  Fe  >  Pb  >  [H]  >  Cu  >  Hg  >  Ag  >  Pt  >  Au
<---------------- High Reactivity ---------------->   <-------------- Noble Metals -----------
Displaces H₂ from cold water                          Displaces H₂ from steam   Cannot displace
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1. Reaction with Water

- **Cold Water**: Only the most electropositive alkali and alkaline earth metals (K, Na, Ca) react with cold water:

$$2Na + 2H_2O \rightarrow 2NaOH + H_2 \uparrow + \text{heat}$$

  Sodium and Potassium float, melt into silvery spheres, and skitter across the surface; the liberated H_2 gas catches fire with explosive popping. Calcium reacts less violently, and the clinging H_2 bubbles cause the metal to float.
- **Hot Water**: Magnesium reacts with hot water to form $Mg(OH)_2$ and H_2.
- **Steam**: Aluminium, Zinc, and Iron do not react with cold or hot liquid water, but react with steam at red heat:

$$2Al + 3H_2O\ (\text{steam}) \rightarrow Al_2O_3 + 3H_2 \uparrow$$

$$3Fe + 4H_2O\ (\text{steam}) \rightleftharpoons Fe_3O_4 + 4H_2 \uparrow$$

- **Unreactive**: Pb, Cu, Hg, Ag, Au do not react with water or steam at any temperature.

2. Reaction with Acids

- Metals situated above Hydrogen in the reactivity series displace hydrogen gas from dilute hydrochloric and sulphuric acids:

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2 \uparrow$$

$$Fe + H_2SO_4 \rightarrow FeSO_4 + H_2 \uparrow$$

- **Exception for Nitric Acid (HNO_3)**: Concentrated and moderately dilute HNO_3 is an aggressive oxidizing agent. It oxidizes evolved hydrogen immediately into water, reducing its own nitrogen species:

$$3Cu + 8HNO_3\ (\text{dilute}) \rightarrow 3Cu(NO_3)_2 + 2NO\ (\text{Nitric oxide}) + 4H_2O$$

$$Cu + 4HNO_3\ (\text{conc.}) \rightarrow Cu(NO_3)_2 + 2NO_2\ (\text{Nitrogen dioxide}) + 2H_2O$$

  Only **Magnesium (Mg)** and **Manganese (Mn)** react with very dilute (1%) cold HNO_3 to evolve pure H_2 gas:

$$Mg + 2HNO_3\ (1\%\ \text{dilute}) \rightarrow Mg(NO_3)_2 + H_2 \uparrow$$

3. Aqua Regia Chemistry

- Aqua Regia is a mixture of **3 parts concentrated HCl and 1 part concentrated HNO_3** by volume ($3:1$ ratio).
- **Mechanism of Dissolution**:

$$HNO_3 + 3HCl \rightarrow NOCl\ (\text{Nitrosyl chloride}) + Cl_2\ (\text{free nascent chlorine}) + 2H_2O$$

The highly reactive nascent chlorine attacks noble metals that resist all individual acids, forming soluble coordinate complexes:

$$Au + 3[Cl] + HCl \rightarrow H[AuCl_4] \text{ (Chloroauric acid)}$$

$$Pt + 4[Cl] + 2HCl \rightarrow H_2[PtCl_6] \text{ (Chloroplatinic acid)}$$

3.4 Important Ores, Extraction Processes & Metallurgy

1. Extractive Steps in Metallurgy

1. **Crushing and Pulverization**: Mechanical reduction of large rocky boulders into fine powder using jaw crushers and ball mills.
2. **Concentration / Beneficiation**: Removal of unwanted rocky, earthy gangue impurities: - *Gravity Separation (Hydraulic Washing)*: Relies on density differences (used for heavy oxide ores like haematite Fe_2O_3). - *Magnetic Separation*: Uses electromagnetic rollers (separates magnetic wolframite from non-magnetic tin stone SnO_2). - *Froth Flotation*: Used selectively for **sulphide ores** ($ZnS, PbS, CuFeS_2$). Pulverized ore is mixed with water, pine oil (collector), and sodium ethyl xanthate. Air is vigorously bubbled through; oil-coated sulphide particles adhere to the rising froth, while hydrophilic gangue sinks to the tank bottom. - *Chemical Leaching*: Selective dissolution of metal in a chemical reagent (e.g., Bayer's process for bauxite using hot concentrated $NaOH$).
3. **Conversion to Oxide (Calcination vs. Roasting)**: - *Calcination*: Heating carbonate or hydrated ores below melting point in the **absence/deficiency of air** to expel moisture and CO_2 ($CaCO_3 \rightarrow CaO + CO_2$). - *Roasting*: Heating sulphide ores below melting point in **excess air** to oxidize sulphides into oxides and volatile SO_2 gas ($2PbS + 3O_2 \rightarrow 2PbO + 2SO_2$).
4. **Reduction of Metal Oxide to Free Metal**: - *Smelting with Carbon (Coke)*: Used for moderate-reactivity metals (Fe, Zn, Pb, Sn). - *Thermite Reduction (Aluminothermy)*: Reduction of refractory oxides ($Cr_2O_3, Mn_3O_4, Fe_2O_3$) using aluminium powder. Highly exothermic ($> 2500^\circ\text{C}$); used for in-situ railway track welding. - *Electrolytic Reduction*: Used for highly electropositive metals (Na, K, Ca, Mg, Al) whose oxides cannot be reduced by carbon.
5. **Refining / Purification**: - *Liquation*: For low-melting metals like Tin (Sn) and Bismuth (Bi). - *Distillation*: For volatile metals with low boiling points like Zinc (Zn) and Mercury (Hg). - *Electrolytic Refining*: Impure metal forms anode, pure metal strip forms cathode, metal salt is electrolyte (used for Cu, Ag, Au, Al). - *Zone Refining*: Based on the principle that impurities are more soluble in the molten melt than in the solid metal; yields ultra-pure semiconductors (Silicon Si, Germanium Ge). - *Vapour Phase Refining*: Mond process for Nickel ($Ni + 4CO \xrightarrow{50^\circ\text{C}} Ni(CO)_4 \xrightarrow{200^\circ\text{C}} Ni + 4CO$); Van Arkel-de Boer process for Titanium (Ti) and Zirconium (Zr) using iodine.

3.5 Corrosion Mechanisms & Protective Technologies

1. Electrochemical Theory of Rusting

Rusting of iron is not a simple chemical combination, but an **electrochemical cell (galvanic corrosion)** set up on the microscopic metal surface: - **Anodic Region**: Iron metal oxidizes, releasing electrons:

$$\text{Anode: } Fe\,(\text{s}) \rightarrow Fe^{2+}\,(\text{aq}) + 2e^- \quad (E^\circ = -0.44\ \text{V})$$

- **Cathodic Region**: Electrons migrate through the metal to a separate region with dissolved oxygen and moisture:

$$\text{Cathode: } O_2\,(\text{g}) + 4H^+\,(\text{aq}) + 4e^- \rightarrow 2H_2O\,(\text{l}) \quad (E^\circ = +1.23\ \text{V})$$

- **Overall Cell Reaction**:

$$2Fe + O_2 + 4H^+ \rightarrow 2Fe^{2+} + 2H_2O$$

Ferrous ions are further oxidized by atmospheric oxygen to ferric state and precipitate as hydrated ferric oxide:

$$4Fe^{2+} + O_2 + 4H_2O + 2xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O\,(\text{Rust}) + 8H^+$$

- **Weight Consequence**: Because atmospheric oxygen and water molecules are chemically bonded into the crystal structure of the iron object, the **total weight of the iron article increases upon rusting**.

2. Corrosion in Other Metals

- **Copper Patina**: When exposed to moist air containing CO_2 and SO_2 over long periods, copper develops a protective green surface layer consisting of **basic copper carbonate**:

$$2Cu + O_2 + CO_2 + H_2O \rightarrow CuCO_3 \cdot Cu(OH)_2\,(\text{Green Patina})$$

  (Seen on the Statue of Liberty and historic temple domes).
- **Silver Tarnishing**: Silver is inert to oxygen, but reacts with trace hydrogen sulphide (H_2S) in polluted air to form a black surface tarnish of **silver sulphide**:

$$2Ag + H_2S \rightarrow Ag_2S\,(\text{Black tarnish}) + H_2$$

3. Prevention Strategies

- **Galvanization**: Dipping iron articles into a bath of molten Zinc ($\approx 450^\circ\text{C}$). Zinc oxidizes preferentially to form a tough film of zinc hydroxycarbonate. Crucially, because Zinc is more electropositive than Iron ($E^\circ_{Zn^{2+}/Zn} = -0.76\ \text{V}$ vs $E^\circ_{Fe^{2+}/Fe} = -0.44\ \text{V}$), zinc acts as a **sacrificial anode**. Even if the zinc coating is scratched, zinc continues to oxidize and supply electrons to the iron, preventing the iron from rusting.
- **Cathodic Protection (Sacrificial Anode)**: Underground steel pipelines and ship hulls are connected via electrical cables to blocks of more reactive sacrificial metals like **Magnesium (Mg) or Zinc (Zn)**. The sacrificial metal corrodes over time and is periodically replaced, preserving the steel structure indefinitely.
- **Anodizing**: An electrolytic oxidation process applied primarily to aluminium articles. The aluminium part forms the anode in an acid bath (H_2SO_4); oxygen liberated at the anode reacts with aluminium to produce a thick, durable, corrosion-resistant surface layer of **aluminium oxide (Al_2O_3)**, which can be dyed in bright decorative colors.

Bilingual Terminology -- Metals and Chemical Reactions

English Term	Hindi Term	Key Anchor
Metal (धातु)	धातु	Lustrous, malleable, ductile, good conductor; loses electrons
Non-metal (अधातु)	अधातु	Dull, brittle, poor conductor (except graphite); gains electrons
Alloy (मिश्र धातु)	मिश्र धातु	Mixture of two or more metals (or metal + non-metal)
Corrosion (संक्षारण)	संक्षारण	Deterioration of metals by reaction with environment; e.g., rusting
Rusting (जंग लगना)	जंग लगना	Corrosion of iron; Fe2O3.xH2O (hydrated iron oxide); needs O2 + H2O
Oxidation (ऑक्सीकरण)	ऑक्सीकरण	Loss of electrons; increase in oxidation state
Reduction (अपचयन)	अपचयन	Gain of electrons; decrease in oxidation state
Redox Reaction (ऑक्सीकरण-अपचयन)	रेडॉक्स अभिक्रिया	Simultaneous oxidation and reduction in same reaction
Calcination (निस्तापन)	निस्तापन	Heating ore in limited air; e.g., CaCO3 -> CaO + CO2
Roasting (भर्जन)	भर्जन	Heating sulphide ore in excess air; converts to oxide; e.g., ZnS + O2 -> ZnO + SO2
Smelting (प्रगलन)	प्रगलन	Melting ore with reducing agent (coke) to extract metal
Electrolytic Refining (विद्युत-अपघटनी परिष्करण)	विद्युतअपघटनी शोधन	Purification of metals using electrolysis; crude as anode, pure as cathode
Thermite Reaction	थर्माइट अभिक्रिया	Al + Fe2O3 -> Al2O3 + Fe + huge heat; welding railway tracks
Galvanisation (जस्तीकरण)	गैल्वनीकरण	Coating iron with zinc to prevent rusting
Anodising (एनोडाइजिंग)	एनोडाइजिंग	Electrochemical coating of aluminium with Al2O3 (protection layer)
Reactivity Series (क्रियाशीलता श्रृंखला)	अभिक्रियाशीलता क्रम	Metals arranged by reactivity: K>Na>Ca>Mg>Al>Zn>Fe>Ni>Sn>Pb>H>Cu>Hg>Ag>Au>Pt

One-Liner Revision -- Metals and Chemical Reactions

#	Fast Fact
1	Most reactive metal = **Potassium (K)**; reacts violently with water
2	Least reactive metal = **Platinum (Pt)** / Gold (Au) = noble metals
3	Most malleable metal = **Gold (Au)** (1g can be beaten into 1m2 sheet)
4	Best conductor of electricity = **Silver (Ag)**; then Cu, then Au, then Al
5	Hardest metal = **Chromium (Cr)**; hardest element = Carbon (diamond)

#	Fast Fact
6	Lightest metal = **Lithium (Li)**; floats on water and oil
7	Mercury (Hg) = only metal that is liquid at room temperature
8	Gallium (Ga) = melts in palm (melting point 29.76 degC)
9	Stainless steel = Fe + Cr (10.5%+) + Ni; rust-resistant
10	Brass = **Cu + Zn**; Bronze = Cu + Sn; Gunmetal = Cu + Sn + Zn
11	German silver / Nickel silver = Cu + Zn + Ni (no silver!)
12	Solder = **Pb + Sn** (used in electronics for joining); being replaced by Sn-Ag-Cu
13	Bell metal = **Cu + Sn** (high tin content); used for bells and cymbals
14	Duralumin = **Al + Cu + Mg + Mn**; lightweight strong alloy; aircraft construction
15	Rusting requires both **water** and **oxygen**; prevented by galvanisation, painting, oiling
16	Galvanisation = coating iron with **zinc (Zn)**; zinc acts as sacrificial anode
17	Thermite reaction: 2Al + Fe2O3 -> 2Fe + Al2O3 + heat (~2500 degC); railway welding
18	Reactivity series: metals above H displace H from acids; K, Na, Ca react with cold water
19	Gold does not corrode; found native (in free state) in nature
20	Bauxite (Al2O3.2H2O) = ore of aluminium; Haematite (Fe2O3) = ore of iron
21	Bayer Process = extraction of alumina from bauxite; then Hall-Heroult for Al electrolysis
22	Blast furnace = extraction of iron from haematite using coke + limestone
23	Pig iron = crude iron from blast furnace; high carbon (4-5%); brittle
24	Wrought iron = nearly pure iron; very low carbon (<0.02%); most malleable
25	Steel = iron + carbon (0.2-2%); iron + Cr = stainless; iron + Mn = manganese steel
26	Amalgamation = extraction of Ag/Au using Hg (forms amalgam)
27	Electrolytic refining: crude metal at anode (-) dissolves; pure at cathode (-) deposits
28	Patina = green coating on copper = basic copper carbonate [Cu(OH)2.CuCO3]
29	Verdigris (patina on Statue of Liberty) = green due to copper oxidation
30	Chemical change = new substance formed (rust); Physical change = no new substance (melting)